NN
ACADEMY

All of IGCSE Maths in 10 Minutes

Whole-syllabus question deck — Cambridge IGCSE Mathematics 0580, Extended. The 10-minute cut: every topic that actually costs marks, one question per page, worked live on screen. Blank workspace sized for annotation in Xournal++.

39	53 / 72	9:59	16 : 9
questions	Extended subtopics	target runtime	254 × 143 mm

PREPARED FOR	DATE	SYLLABUS	EXAM SERIES
Nur Nabi	**21 September 2026**	**0580 (2025–2027) · Extended**	**2026 & 2027**

READ THIS BEFORE YOU RECORD

How to use this deck

In Xournal++

- Open this PDF as the **background** (Annotate PDF), not as an image import — the text stays crisp at any zoom.
- Page size is **254 × 143 mm**, exactly 16:9, so a full-page view fills a 1920 × 1080 recording frame with no letterboxing.
- The faint dot grid is 5 mm and is there to keep your handwriting level. It is light enough to disappear under video compression.
- Set the pen to roughly **1.5 mm**, dark blue or black. Use red only for the final answer so the freeze-frame reads at a glance.
- Export the annotated file afterwards as the downloadable "worked solutions" PDF — that is your description link and your email-list magnet.

Pacing

- Each page carries its own time budget in the top-right. It is a ceiling, not a target.
- If a page overruns, cut the explanation, never the worked answer.
- Hard cut between pages. No transitions and no "now let's look at" — there is no runtime for connective tissue.

Everything here is original. The questions were written for this deck against the published syllabus. Cambridge past-paper questions are copyright and must not be screen-recorded or retyped — these are fresh questions in the same style, which is why you can publish and monetise the video safely. The whole-syllabus sweep format is a genre, not anyone's property; the wording, the numbers and the working are yours.

Say this at the end, and put it in the description. A 10-minute sweep is a revision aid and a gap-finder, not a course. Students who rely on it alone will be short of practice. Four seconds of honesty buys you the comment section.

SEGMENT MAP

The 10-minute running order

> Paste the **Start** and **Segment** columns into the YouTube description as chapter markers. The **Budget** column is your edit discipline, and at this length there is no slack in it at all.

START	SEGMENT	CODES	BUDGET
0:00	**Intro — the promise**	–	0:30
0:30	**Number** 10 questions	E1.1 - E1.18	2:04
2:34	**Algebra and graphs** 11 questions	E2.1 - E2.13	2:16
4:50	**Coordinate geometry** 1 questions	E3.1 - E3.7	0:24
5:14	**Geometry** 3 questions	E4.1 - E4.8	0:46
6:00	**Mensuration** 3 questions	E5.1 - E5.5	0:42
6:42	**Trigonometry** 4 questions	E6.1 - E6.6	1:02
7:44	**Transformations and vectors** 2 questions	E7.1 - E7.4	0:44
8:28	**Probability** 2 questions	E8.1 - E8.4	0:24
8:52	**Statistics** 3 questions	E9.1 - E9.7	0:42
9:34	**Outro — how to use this**	–	0:25
9:59	**Total runtime**		

1

Number

E1.1 - E1.18 · 10 questions · 2:04 on the clock

Question 01

Types of number, HCF and LCM

Also know: natural numbers · integers · primes · squares · cubes · rational and irrational · reciprocals

(a) Write 168 and 180 each as a product of its prime factors.

..... [2]

(b) Hence find the highest common factor and the lowest common multiple of 168 and 180.

..... [2]

Question 02

Sets and Venn diagrams

In a group of 40 students, 24 study Physics, 19 study Chemistry and 7 study neither.

Notation: ξ $\cup$ $\cap$ A' $\in$ $\notin$ $\subseteq$ $\varnothing$ $n(A)$

(a) Complete the Venn diagram.

..... [2]

(b) Find n($P \cap C$).

..... [1]

Question 02

Sets and Venn diagrams

In a group of 40 students, 24 study Physics, 19 study Chemistry and 7 study neither.

(c) Find n($P \cap C'$).

..... [1]

Question 03

Powers, roots and the index laws

Recall squares to 15^2 and cubes of 1, 2, 3, 4, 5 and 10

(a) Work out $27^{-2/3}$, giving your answer as a fraction.

..... [2]

(b) Simplify $(2a^3b)^4 \div 4a^5b^2$.

..... [2]

Question 04

Fractions, decimals and recurring decimals

(a) Write $0.4\dot{2}$ as a fraction in its simplest form.

..... [2]

(b) Work out $2\frac{1}{3} - 1\frac{3}{4}$, giving your answer as a fraction in its simplest form.

..... [2]

Question 05

Standard form

Answers must be in the form $A \times 10^n$ with $1 \leq A < 10$

(a) Work out $(3.2 \times 10^5) \div (8 \times 10^{-2})$.

..... [2]

(b) Work out $(6 \times 10^4) + (4.5 \times 10^5)$.

..... [2]

Question 06

Limits of accuracy (bounds)

A rectangle measures 12.4 cm by 8.6 cm, each correct to 1 decimal place.

(a) Write down the lower and upper bound of each side.

..... [2]

(b) Calculate the upper bound of the area.

..... [2]

(c) Calculate the lower bound of the perimeter.

..... [2]

Question 07

Ratio and proportion

An amount of money is shared between Amina, Bilal and Chowdhury in the ratio 3 : 5 : 7.

Chowdhury receives $84 more than Amina. Find the total amount of money shared.

WORKING

Question 08

Percentages and reverse percentages

(a) After a 15% increase, the price of a phone is $161. Calculate the price before the increase.

..... [2]

(b) $2000 is invested at 3.5% per year compound interest. Find the value after 5 years.

..... [3]

Question 09

Exponential growth and decay

A car is bought for $18 000. Its value depreciates by 12% each year.

Calculate its value after 4 years, correct to the nearest dollar.

Growth: $P(1 + r)^n$ · Decay: $P(1 - r)^n$ · knowledge of e is not required

WORKING

Question 10

Surds

(a) Simplify √50 + √18.

..... [2]

(b) Simplify 6 ÷ √3.

..... [2]

(c) Rationalise the denominator of 1 ÷ (2 + √3).

..... [2]

2

Algebra and graphs

E2.1 - E2.13 · 11 questions · 2:16 on the clock

Question 11

Expanding and factorising

(a) Expand and simplify $(2x - 3)(x + 5)$.

[2]

(b) Factorise $6x^2 - x - 12$.

[2]

Question 11

Expanding and factorising

(c) Factorise $9x^2 - 49y^2$.

..... [2]

(d) Factorise by grouping: $ax + ay - 2bx - 2by$.

..... [2]

Question 12

Algebraic fractions

(a) Simplify $(x^2 - 9) \div (x^2 + 7x + 12)$.

[3]

(b) Write as a single fraction $3/(x + 1) + 2/(x - 2)$.

[3]

Question 13

Equations and rearranging

(a) Solve simultaneously $3x + 2y = 16$ and $x - y = 2$.

..... [3]

(b) Solve $2x^2 - 7x - 3 = 0$, giving answers correct to 2 decimal places.

..... [4]

Question 13

Equations and rearranging

(c) Make r the subject of $V = \frac{2}{3}\pi r^3$.

..... [3]

(d) Write $x^2 - 6x + 2$ in the form $(x - a)^2 - b$.

..... [2]

Question 14

Non-linear simultaneous equations

Solve simultaneously

$$y = x^2 - 4 \quad \text{and} \quad y = 3x$$

WORKING

Question 15

Inequalities

Dividing by a negative reverses the inequality sign

(a) Solve $5 - 2x < 11$ and show the solution on a number line.

..... [3]

(b) Solve $-3 \leq 2x + 1 < 9$.

..... [2]

Question 16

Sequences

Find the *n*th term of each sequence.

Linear · quadratic · cubic · exponential · and simple combinations

(a) 7, 11, 15, 19, ...

..... [2]

(b) 3, 8, 15, 24, 35, ...

..... [2]

Question 16

Sequences

Find the *n*th term of each sequence.

(c) 5, 10, 20, 40, ...

..... [2]

Question 17

Direct and inverse proportion

y is inversely proportional to x^2. When $x = 2$, $y = 12$.

(a) Find y in terms of x.

..... [2]

(b) Find y when $x = 4$.

..... [1]

(c) Find x when $y = 0.75$.

..... [2]

Question 18

Graphs in practical situations

A car accelerates uniformly from rest to 15 m/s in 10 s, travels at 15 m/s for 30 s, then decelerates uniformly to rest in 20 s.

(a) Sketch the speed–time graph.

[2]

(b) Find the acceleration in the first 10 s.

[1]

(c) Find the total distance travelled.

[3]

Question 19

Sketching curves

Know the shapes: linear · quadratic · cubic · reciprocal · exponential

(a) Write $y = x^2 - 6x + 5$ in completed-square form and state the turning point and the roots.

[4]

(b) Sketch $y = 4/x$ and state the equations of its asymptotes.

[3]

Question 20

Differentiation

$y = x^3 - 6x^2 + 5$

(a) Find dy/dx.

..... [2]

(b) Find the coordinates of the two stationary points.

..... [4]

(c) Determine which is a maximum and which is a minimum.

..... [3]

Question 21

Functions

$f(x) = 3x - 4 \quad g(x) = x^2$

(a) Find fg(2).

..... [1]

(b) Find gf(2).

..... [1]

Question 21

Functions

$f(x) = 3x - 4 \quad g(x) = x^2$

(c) Find $f^{-1}(x)$.

..... [2]

(d) Find $ff(x)$ in its simplest form.

..... [2]

3

Coordinate geometry

E3.1 - E3.7 · 1 questions · 0:24 on the clock

Question 22

All of coordinate geometry in one pair of points

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

One pair of points answers all seven subtopics — that is the trick that makes this segment fit

(a) Midpoint of AB.

..... [1]

(b) Length of AB.

..... [2]

Question 22

All of coordinate geometry in one pair of points

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

(c) Gradient of AB.

..... [1]

(d) Equation of AB.

..... [2]

Question 22

All of coordinate geometry in one pair of points

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

(e) Equation of the line through the origin parallel to AB.

..... [1]

(f) Equation of the line through B perpendicular to AB.

..... [3]

4

Geometry

E4.1 - E4.8 · 3 questions · 0:46 on the clock

Question 23

Geometrical terms and angles

(a) For a regular 12-sided polygon, find the size of each exterior angle, each interior angle, and the sum of the interior angles.

..... [3]

(b) Two parallel lines are cut by a transversal. Name the angle relationships you would use and give a reason for each.

..... [3]

Question 24

Similarity and symmetry

Two similar cones have heights 4 cm and 6 cm. The smaller cone has volume 32 cm^3 and surface area 20 cm^2.

Length factor k → area factor k^2 → volume factor k^3

(a) Find the volume of the larger cone.

..... [2]

(b) Find its surface area.

..... [2]

Question 25

Circle theorems

Find the lettered angles, giving a geometrical reason for each.

Semicircle · tangent ⊥ radius · centre = 2 × circumference · same segment · cyclic quadrilateral · alternate segment · equal chords · tangents from a point are equal

(a) Angle at the centre is 130° — find the angle at the circumference.

..... [2]

(b) Hence the opposite angle of the cyclic quadrilateral.

..... [2]

Question 25

Circle theorems

Find the lettered angles, giving a geometrical reason for each.

(c) A tangent meets a chord at 58° — find the angle in the alternate segment.

..... [1]

(d) *OP* = 17 cm, radius 8 cm — find the tangent length *PT*.

..... [3]

5

Mensuration

E5.1 - E5.5 · 3 questions · 0:42 on the clock

Question 26

Area, perimeter and compound shapes

(a) A trapezium has parallel sides 7 cm and 11 cm and perpendicular height 6 cm. Find its area.

..... [2]

(b) A rectangle 10 cm by 6 cm has a semicircle of diameter 6 cm removed from one end. Find the area remaining.

..... [3]

Question 27

Circles, arcs and sectors

A sector of a circle has radius 9 cm and angle 140°.

(a) Find the arc length.

..... [2]

(b) Find the area of the sector.

..... [2]

(c) Find the perimeter of the sector.

..... [1]

Question 28

Surface area and volume

Cone: $\frac{1}{3}\pi r^2 h$ and curved area $\pi r l$ · Sphere: $\frac{4}{3}\pi r^3$ and $4\pi r^2$ — both given in the formula list

(a) A cone has radius 5 cm and vertical height 12 cm. Find its slant height, its total surface area and its volume.

[5]

(b) A sphere has radius 6 cm. Find its volume and its surface area.

[3]

6

Trigonometry

E6.1 - E6.6 · 4 questions · 1:02 on the clock

Question 29

Pythagoras and right-angled trigonometry

In a right-angled triangle the side opposite angle θ is 7 cm and the hypotenuse is 12 cm.

SOH · CAH · TOA

(a) Find θ.

..... [2]

(b) Find the third side.

..... [2]

Question 30

Trigonometric graphs and equations

Solve for $0° \leq x \leq 360°$.

Sketch the curve first — it tells you how many solutions there are

(a) sin x = 0.5

..... [2]

(b) 2 cos x + 1 = 0

..... [3]

Question 31

Sine rule, cosine rule and area

Two sides and the angle between → cosine rule. Anything else → sine rule. Both are in the formula list.

(a) In triangle ABC, $A = 47°$, $B = 63°$, $a = 8.4$ cm. Find b.

[3]

(b) In a different triangle, $b = 8$ cm, $c = 11$ cm, $A = 47°$. Find a.

[3]

(c) Find its area.

[2]

Question 32

Pythagoras and trigonometry in 3D

A cuboid has a rectangular base 6 cm by 8 cm and height 10 cm.

(a) Find the length of the space diagonal.

..... [2]

(b) Find the angle between the space diagonal and the base.

..... [3]

7

Transformations and vectors

E7.1 - E7.4 · 2 questions · 0:44 on the clock

Question 33

Transformations

Describe each transformation fully.

Reflection: the mirror line · Rotation: angle, direction, centre · Enlargement: scale factor, centre · Translation: the column vector

(a) A reflection.

..... [2]

(b) A rotation.

..... [3]

Question 33

Transformations

Describe each transformation fully.

(c) An enlargement with scale factor −2, centre (1, 1).

..... [3]

(d) A translation.

..... [2]

Question 34

Vectors, magnitude and vector geometry

$\mathbf{a} = (3, -1)$ and $\mathbf{b} = (-2, 5)$ as column vectors.

(a) Find $2\mathbf{a} + \mathbf{b}$.

..... [2]

(b) Find $|2\mathbf{a} + \mathbf{b}|$.

..... [2]

(c) $OA = \mathbf{a}$, $OB = \mathbf{b}$, M is the midpoint of AB. Find AB and OM in terms of $\mathbf{a}$ and $\mathbf{b}$.

..... [3]

8

Probability

E8.1 - E8.4 · 2 questions · 0:24 on the clock

Question 35

Relative and expected frequency

The probability that a machine produces a faulty component is 0.15.

(a) Find the expected number of faulty components in a batch of 400.

..... [1]

(b) In a trial, 72 out of 400 were faulty. Find the relative frequency and comment.

..... [2]

Question 36

Tree diagrams and conditional probability

A bag holds 5 red and 3 blue counters. Two are taken at random **without replacement**.

*AND → multiply along branches · OR → add the outcomes · without replacement changes **both** numerator and denominator*

(a) P(both red).

[2]

(b) P(one of each colour).

[3]

Question 36

Tree diagrams and conditional probability

A bag holds 5 red and 3 blue counters. Two are taken at random **without replacement**.

(c) P(second is red, given the first was red).

..... [1]

9

Statistics

E9.1 - E9.7 · 3 questions · 0:42 on the clock

Question 37

Averages and measures of spread

The table shows the time t minutes taken by 30 students to finish a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	4	11	9	6

(a) Calculate an estimate of the mean.

..... [4]

(b) Write down the modal class.

..... [1]

Question 37

Averages and measures of spread

The table shows the time t minutes taken by 30 students to finish a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	4	11	9	6

(c) Identify the class containing the median.

..... [1]

Question 38

Cumulative frequency diagrams

A cumulative frequency curve is drawn for 80 values.

*Plot points at the **upper** class boundary and join with a smooth curve*

(a) At what value of the cumulative frequency do you read the median, the lower quartile and the upper quartile?

..... [3]

(b) If LQ = 24 and UQ = 41, find the interquartile range.

..... [1]

Question 39

Histograms

A histogram has unequal class widths.

*frequency density = frequency ÷ class width — the vertical axis is **always** labelled Frequency density*

(a) The class $20 < x \leq 50$ has frequency 24. Find the frequency density.

..... [2]

(b) Another bar has frequency density 2.4 over the class $50 < x \leq 60$. Find its frequency.

..... [2]

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Answer key 1 of 4

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
01	**E1.1**	Types of number, HCF and LCM	$168 = 2^3 \times 3 \times 7$ $180 = 2^2 \times 3^2 \times 5$ | HCF $= 2^2 \times 3 =$ **12** LCM $= 2^3 \times 3^2 \times 5 \times 7 =$ **2520**
02	**E1.2**	Sets and Venn diagrams	$n(P \cup C) = 40 - 7 = 33$, so $24 + 19 - x = 33 \rightarrow$ **$n(P \cap C) = 10$**; $n(P \cap C') = 24 - 10 =$ **14**
03	**E1.3 / E1.7**	Powers, roots and the index laws	(a) $27^{-2/3} = 1 \div (\sqrt[3]{27})^2 = 1 \div 9 =$ **1/9** | (b) $16a^{12}b^4 \div 4a^5b^2 =$ **$4a^7b^2$**
04	**E1.4**	Fractions, decimals and recurring decimals	(a) $0.4\dot{2} = 42/99 =$ **14/33** | (b) $7/3 - 7/4 = (28 - 21)/12 =$ **7/12**
05	**E1.8**	Standard form	(a) $0.4 \times 10^7 =$ **4×10^6** | (b) $60\,000 + 450\,000 = 510\,000 =$ **5.1×10^5**
06	**E1.10**	Limits of accuracy (bounds)	$12.35 \le L < 12.45$, $8.55 \le W < 8.65$ | (b) $12.45 \times 8.65 =$ **107.6925 cm²** | (c) $2(12.35 + 8.55) =$ **41.8 cm**
07	**E1.11**	Ratio and proportion	$7 - 3 = 4$ parts = \$84, so 1 part = \$21. Total $= 15 \times 21 =$ **\$315**
08	**E1.13**	Percentages and reverse percentages	(a) $161 \div 1.15 =$ **\$140** (not 161×0.85) | (b) $2000 \times 1.035^5 =$ **\$2375.37**
09	**E1.17**	Exponential growth and decay	$18\,000 \times 0.88^4 =$ **\$10 795** (10 794.55)
10	**E1.18**	Surds	(a) $5\sqrt{2} + 3\sqrt{2} =$ **$8\sqrt{2}$** | (b) $6\sqrt{3}/3 =$ **$2\sqrt{3}$** | (c) $\times (2 - \sqrt{3})/(2 - \sqrt{3}) \rightarrow (2 - \sqrt{3})/1 =$ **$2 - \sqrt{3}$**

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Answer key 2 of 4

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
11	E2.2	Expanding and factorising	(a) **$2x^2 + 7x - 15$** | (b) **$(2x - 3)(3x + 4)$** | (c) **$(3x - 7y)(3x + 7y)$** | (d) **$(a - 2b)(x + y)$**
12	E2.3	Algebraic fractions	(a) $(x - 3)(x + 3) / (x + 3)(x + 4) =$ **$(x - 3)/(x + 4)$** | (b) **$(5x - 4) / (x + 1)(x - 2)$**
13	E2.5	Equations and rearranging	(a) **$x = 4, y = 2$** | (b) $x = (7 \pm \sqrt{73})/4 =$ **3.89 or −0.39** | (c) **$r = \sqrt[3]{(3V / 4\pi)}$** | (d) **$(x - 3)^2 - 7$**
14	E2.5	Non-linear simultaneous equations	$x^2 - 3x - 4 = 0 \rightarrow (x - 4)(x + 1) = 0 \rightarrow$ **$x = 4, y = 12$** or **$x = -1, y = -3$**
15	E2.6	Inequalities	(a) $-2x < 6 \rightarrow$ **$x > -3$** (open circle) | (b) $-4 \le 2x < 8 \rightarrow$ **$-2 \le x < 4$**
16	E2.7	Sequences	(a) **$4n + 3$** | (b) second difference 2 → **$n^2 + 2n$** | (c) **$5 \times 2^{n-1}$**
17	E2.8	Direct and inverse proportion	(a) **$y = 48/x^2$** | (b) **$y = 3$** | (c) $x^2 = 64 \rightarrow$ **$x = 8$**
18	E2.9	Graphs in practical situations	(b) $15 \div 10 =$ **1.5 m/s²** | (c) area = ½(10)(15) + 30(15) + ½(20)(15) = 75 + 450 + 150 = **675 m**
19	E2.11	Sketching curves	(a) $(x - 3)^2 - 4$, turning point **(3, −4)**, roots **$x = 1$ and $x = 5$** | (b) asymptotes **$x = 0$ and $y = 0$**
20	E2.12	Differentiation	(a) **$3x^2 - 12x$** | (b) $3x(x - 4) = 0 \rightarrow$ **(0, 5)** and **(4, −27)** | (c) $d^2y/dx^2 = 6x - 12$: at $x = 0$ it is $-12 \rightarrow$ **maximum**; at $x = 4$ it is $12 \rightarrow$ **minimum**

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Answer key 3 of 4

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
21	E2.13	Functions	(a) f(4) = **8** | (b) g(2) = **4** | (c) **(x + 4)/3** | (d) 3(3x − 4) − 4 = **9x − 16**
22	E3.1 - E3.7	All of coordinate geometry in one pair of points	(a) **(2, 4)** | (b) √(8²+6²) = **10** | (c) **¾** | (d) **y = ¾x + 2.5** | (e) **y = ¾x** | (f) gradient −4/3 → **y = −(4/3)x + 15**
23	E4.1 / E4.6	Geometrical terms and angles	(a) exterior = 360 ÷ 12 = **30°**; interior = **150°**; sum = (12 − 2) × 180 = **1800°** | (b) corresponding, alternate, co-interior (sum 180°)
24	E4.4 / E4.5	Similarity and symmetry	k = 1.5 | (a) 32 × 1.5³ = 32 × 3.375 = **108 cm³** | (b) 20 × 1.5² = **45 cm²**
25	E4.7 / E4.8	Circle theorems	(a) **65°** (angle at centre is twice the angle at the circumference) | (b) **115°** (cyclic quadrilateral) | (c) **58°** (alternate segment) | (d) √(289 − 64) = **15 cm** (tangent ⊥ radius)
26	E5.2 / E5.5	Area, perimeter and compound shapes	(a) ½(7 + 11)(6) = **54 cm²** | (b) 60 − ½π(3)² = 60 − 14.14 = **45.9 cm²**
27	E5.3	Circles, arcs and sectors	(a) (140/360)(2π × 9) = **22.0 cm** | (b) (140/360)(π × 81) = **99.0 cm²** | (c) 21.99 + 9 + 9 = **40.0 cm**
28	E5.4	Surface area and volume	(a) l = 13 cm; total SA = π(5)(13) + π(5)² = 90π = **283 cm²**; V = ⅓π(25)(12) = 100π = **314 cm³** | (b) V = 288π = **905 cm³**; SA = 144π = **452 cm²**
29	E6.1 / E6.2	Pythagoras and right-angled trigonometry	(a) sin θ = 7/12 → **θ = 35.7°** | (b) √(144 − 49) = √95 = **9.75 cm**
30	E6.4	Trigonometric graphs and equations	(a) **x = 30° and 150°** | (b) cos x = −0.5 → **x = 120° and 240°**

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Answer key 4 of 4

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
31	E6.5	Sine rule, cosine rule and area	(a) 8.4 sin63° ÷ sin47° = **10.2 cm** | (b) a^2 = 64 + 121 − 176 cos47° = 64.97 → **8.06 cm** | (c) ½(8)(11) sin47° = **32.2 cm²**
32	E6.6	Pythagoras and trigonometry in 3D	(a) √(36 + 64 + 100) = √200 = **14.1 cm** | (b) base diagonal = 10, tan θ = 10/10 → **θ = 45°**
33	E7.1	Transformations	A negative scale factor puts the image on the **opposite side of the centre and inverted**; a fractional scale factor makes it **smaller**. One word per transformation — naming two transformations loses the mark.
34	E7.2 - E7.4	Vectors, magnitude and vector geometry	(a) **(4, 3)** | (b) √(16+9) = **5** | (c) *AB* = **b − a**; *OM* = **a** + ½(**b** − **a**) = **½(a + b)**
35	E8.2	Relative and expected frequency	(a) 0.15 × 400 = **60** | (b) 72/400 = **0.18** — higher than expected, though a single trial is weak evidence
36	E8.3 / E8.4	Tree diagrams and conditional probability	(a) 5/8 × 4/7 = **5/14** | (b) 2 × (5/8 × 3/7) = **15/28** | (c) **4/7** — the condition changes the denominator too
37	E9.3	Averages and measures of spread	(a) Σfx = 20 + 165 + 225 + 210 = 620; 620 ÷ 30 = **20.7 min** | (b) **10 < t ≤ 20** | (c) 15th/16th value → **20 < t ≤ 30**
38	E9.6	Cumulative frequency diagrams	(a) median at **40**, LQ at **20**, UQ at **60** | (b) IQR = 41 − 24 = **17**
39	E9.7	Histograms	(a) 24 ÷ 30 = **0.8** | (b) 2.4 × 10 = **24**